Matrix Theory: Normal Form (Canonical Form)

This study note explains the **Normal Form** of a matrix, which is the most simplified state a matrix can reach using both row and column transformations.

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1. What is Normal Form?

If a matrix A has a rank r, it can always be reduced to a specific block-style structure containing an Identity (Unit) matrix I_r surrounded by zero matrices. This is known as the **Normal** or **Canonical Form**.

The Four Possible Structures

Depending on the original dimensions of the matrix, it will take one of these forms:

Unit matrix with zero rows and columns:

$$\begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$$

Unit matrix with zero rows below:

$$\begin{bmatrix} I_r \\ O \end{bmatrix}$$

Unit matrix with zero columns to the right:

$$\begin{bmatrix} I_r & O \end{bmatrix}$$

The entire matrix reduces to a Unit matrix:

$$\begin{bmatrix} I_r \end{bmatrix}$$

> **Key Rule:** The order of the resulting Identity matrix I is exactly equal to the **Rank** of the original matrix.

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2. Visual Examples (using I_3)

As shown in the lecture, if the rank of a matrix is **3**, the Normal Form might look like any of these:

Type (i):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_3 & O \\ O & O \end{bmatrix}$$

Type (ii):

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_3 \\ O \end{bmatrix}$$

Type (iii):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} I_3 & O \end{bmatrix}$$

Type (iv):

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_3 \end{bmatrix}$$

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3. Step-by-Step Example: Finding Rank via Reduction

To reach Normal Form, we use **Elementary Transformations**. Unlike Echelon form, here we use both Rows (R) and Columns (C).

The Problem

Find the rank of matrix A:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

The Transformation Algorithm

Step 1: Eliminate elements in the first column.
Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

Step 2: Eliminate elements in the second column (Row Operation).
Apply $R_3 \to R_3 - R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

*At this stage, the matrix is in **Triangular (Echelon) Form**.*

Step 3: Further reduction to reach Normal Form.
By continuing with column operations (e.g., $C_2 \to C_2 - 2C_1$) and scaling rows, we eventually reach:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & O \\ O & O \end{bmatrix}$$

Final Result

- **Normal Form:**

$$\begin{bmatrix} I_2 & O \\ O & O \end{bmatrix}$$

- **Rank:** $\rho(A) = 2$ (Since the identity matrix is of order 2).

4. Summary Table for GitHub

Form	Purpose	Transformations Used
Echelon Form	Solving $Ax = B$	Rows Only
Normal Form	Simplest definition of Rank	Rows & Columns